A Novel Variational-Mode-Decomposition-Based Long Short-Term Memory for Foreign Exchange Prediction

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Abstract

The global foreign exchange (forex) market has a daily volume on the order of $5 trillion and is the largest financial market in the world. The importance of this market has attracted many research efforts. Numerous techniques have been developed, including technical analysis, where researchers attempt to predict forex rates based on past forex data. In this paper, we combine a signal processing technique Variational Mode Decomposition (VMD) and the Long Short-Term Memory Neural Network (LSTM) to predict forex and show significant improvements over other predicting models.

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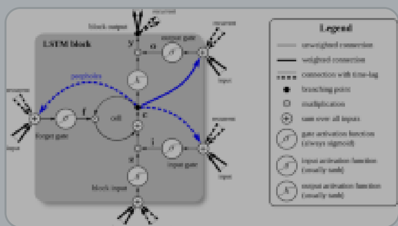

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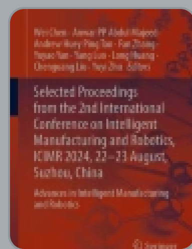

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